



Performance Indicators for Energy Storage in Malaysia's Energy Transition: A Systematic Review and Sustainable Supply Chain Management Framework

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Abstract: Energy storage systems (ESS) play a central role in renewable energy integration, grid reliability, and the transition toward low-carbon energy systems. In Malaysia, however, the indicators used to evaluate ESS remain fragmented, limiting comparison across technologies and weakening the evidence available for investment, policy, and sustainable supply chain decisions. This study investigates how ESS performance has been evaluated in the Malaysian energy transition and develops a structured framework for linking engineering performance with sustainable supply chain management (SSCM). A systematic review of 40 eligible studies was conducted using bibliometric mapping and thematic analysis. The reported indicators were identified, coded, and classified into technical, economic, operational, and policy/environmental dimensions. The results showed that capacity and sizing were the most frequently reported indicators, followed by renewable energy integration and system reliability or availability. Battery-based systems dominated the reviewed literature, particularly in photovoltaic (PV)-coupled applications, whereas long-duration storage, grid-scale services, lifecycle assessment, and end-of-life considerations received limited attention. Although levelized cost of energy (LCOE) and net present cost (NPC) were commonly reported, none of the retained studies explicitly evaluated the levelized cost of storage (LCOS). The findings indicate that current assessment practices remain concentrated on project-level technical and financial performance and provide insufficient support for evaluating material sourcing, lifecycle impacts, regulatory conditions, and supply chain resilience. The proposed framework connects ESS performance evaluation with technology selection, investment appraisal, supplier assessment, environmental management, and policy planning. It provides a systematic basis for developing national performance benchmarks and supports more consistent ESS decision-making in Malaysia and other Association of Southeast Asian Nations (ASEAN) energy systems.

Keywords: Energy storage systems; Performance indicators; Sustainable supply chain management; Energy transition; Systematic review; Malaysia

1 Introduction

Malaysia is navigating a critical transition towards a low-carbon energy system, supported by ambitious policy targets that include achieving a 40% renewable energy share in installed capacity by 2035, as outlined in the Malaysia Renewable Energy Roadmap (MyRER) [1]. This shift is reinforced by the National Energy Transition Roadmap (NETR), which seeks to phase out coal-fired generation and accelerate cleaner alternatives to sustain energy security [2]. Against this background, energy storage systems (ESS) are increasingly recognised as critical enablers of renewable energy adoption, as they mitigate the intermittency of solar and wind power and ensure system reliability as renewable penetration grows [3, 4].

Despite this growing momentum, the Malaysian discourse on ESS remains fragmented. Comparative assessments

of electrical energy storage technologies have also highlighted the importance of lifecycle cost and technology-specific economic indicators in evaluating storage alternatives [5]. Critical performance dimensions such as levelized cost of storage (LCOS), round-trip efficiency (RTE), degradation rates and ancillary service contributions are seldom reported. The absence of such indicators limits comparability across technologies and undermines evidence-based investment and policy decision-making.

Moreover, while much of the current discussion centres on technical and economic aspects, energy storage must also be viewed through the lens of sustainable supply chain management (SSCM). ESS technologies depend on critical raw materials, have complex lifecycles, and pose challenges for recycling and end-of-life management [6, 7]. Integrating ESS evaluation within SSCM frameworks ensures that performance assessments address not only efficiency and costs but also sustainability, resilience, supplier selection, lifecycle considerations, and compliance with environmental standards. Recent engineering management studies further demonstrate that multi-criteria supplier evaluation provides a systematic basis for balancing economic, environmental, and logistical factors throughout sustainable supply chains [8].

This study contributes to addressing the identified gaps through a systematic review of 40 carefully screened publications, combining bibliometric and thematic analyses to evaluate the performance indicators reported in the literature and categorising the identified indicators into four dimensions: technical, economic, operational, and policy/environmental.

2 Literature Review

2.1 Global Trends in Energy Storage

Energy storage has emerged as a key technology in the transition towards low-carbon power systems. The enhanced deployment of renewable energy, particularly solar photovoltaic (PV) systems and wind, has enlarged the urgency of integrating energy storage to balance intermittency, improve grid stability, and afford supplementary services [9]. Global energy storage deployment has expanded rapidly alongside the increasing integration of renewable energy, with battery-based storage playing an increasingly important role in power-system flexibility and energy-transition strategies [10]. Battery-based systems have gained prominence due to their high efficiency, modularity, and declining cost trajectory. Figure 1 illustrates the global installed energy storage capacity under the Stated Policies Scenario (STEPS) and Net Zero Emissions (NZE) Scenario, highlighting the projected rapid expansion of utility-scale battery storage and pumped hydro systems toward 2030 [10].

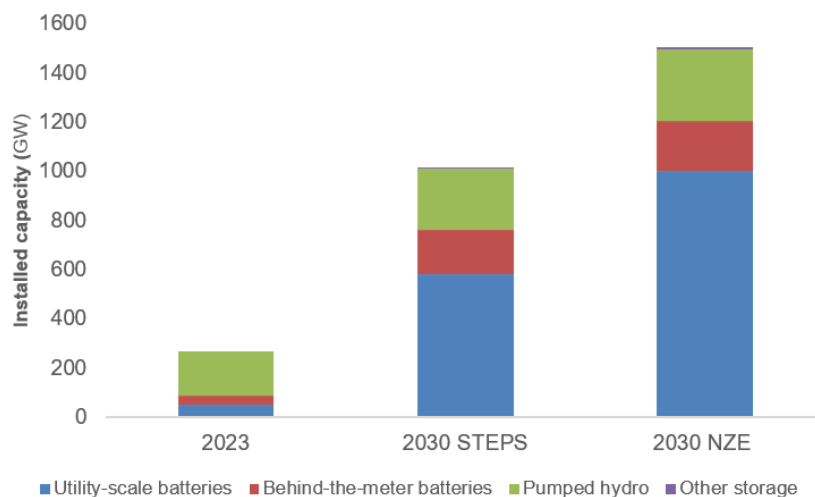


Figure 1. Global installed energy storage capacity by technology under the Stated Policies Scenario (STEPS) and Net Zero Emissions (NZE) Scenario, comparing installed capacity in 2023 with projected deployment in 2030

Source: Adapted from Ref. [10] under the Creative Commons CC BY 4.0 license.

Cost decrease has been one of the most significant global trends. The price of lithium-ion batteries has dropped by nearly 90% since 2010, from above USD 1,200/kWh to around USD 150/kWh in 2022, making large-scale applications economically feasible [11]. Concurrently, advancements in alternate chemistries, such as lithium iron phosphate (LFP) and nickel manganese cobalt (NMC), have extended the operational range, safety, and lifecycle of storage systems. Similarly, vanadium redox flow batteries (VRFBs) are gaining traction for long-term applications, while pumped hydro storage (PHS) remains the largest contributor to installed global capacity, representing almost 70% of total storage volume [3].

Geographically, deployment is concentrated in a few leading markets. China has rapidly scaled its storage fleet, targeting more than 100 GW of new non-PHS capacity by 2030, supported by aggressive industrial policies and domestic manufacturing capabilities. The United States is expanding through market-driven mechanisms, with storage increasingly coupled to utility-scale solar and wind projects under independent system operator (ISO) markets. The European Union promotes storage within its Green Deal framework, with emphasis on cross-border grid integration and energy security. Meanwhile, Japan and South Korea remain pioneers in technology development and safety standards, particularly in advanced batteries and hydrogen-based storage. Figure 2 presents selected indicators of global battery energy storage system (BESS) deployment in 2024 [12]. Approximately 200 GWh of new BESS capacity was added worldwide during the year, with China contributing more than 100 GWh and the United States approximately 35 GWh. By the end of 2024, total global operational BESS capacity had reached approximately 375 GWh. These figures highlight the rapid expansion of battery storage deployment and the substantial contribution of China and the United States to new global capacity additions.

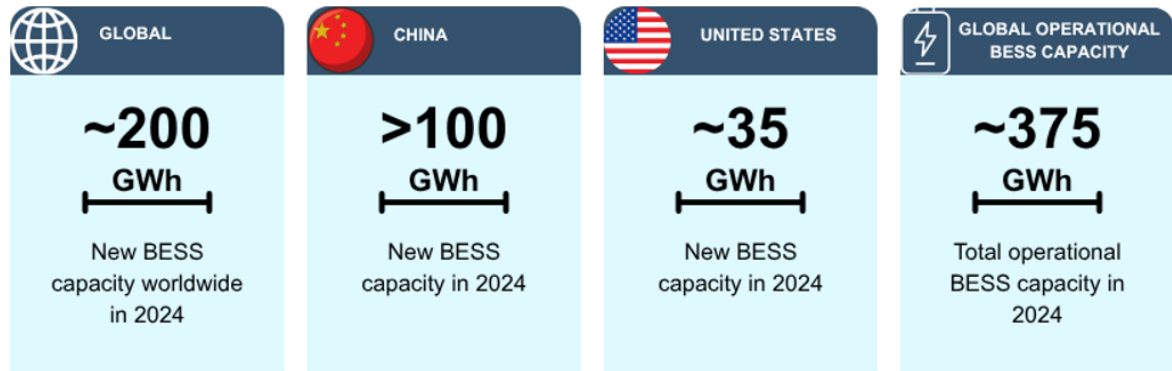


Figure 2. Key indicators of global battery energy storage system (BESS) deployment in 2024
Source: Authors’ reconstruction using data reported by Rystad Energy [12].

Beyond technological and economic dimensions, global scholarship highlights the importance of policy and market design. The availability of ancillary service markets, time-of-use pricing, and capacity remuneration mechanisms strongly influence storage adoption. In contrast, regions lacking such regulatory frameworks face slower uptake despite falling technology costs [10]. Environmental and social concerns, particularly around lifecycle impacts, resource extraction (lithium, cobalt, vanadium), and recycling infrastructure, are increasingly shaping international research and industrial practices.

In summary, global energy storage trends demonstrate rapid expansion, falling costs, and diversification of technologies, motivated by both market incentives and policy mandates. Yet, the distribution of adoption remains uneven, concentrated in technologically advanced economies. These international developments provide valuable benchmarks for assessing Malaysia’s progress and identifying opportunities to accelerate deployment in line with regional and global best practices.

2.2 Renewable Energy in Malaysia

Renewable energy (RE) in Malaysia comprises solar, biomass, hydro, and wind, with solar PV positioned as the central pillar of national energy diversification strategies. Government programmes such as the Large-Scale Solar (LSS) initiative, the Feed-in Tariff mechanism, and the Net Energy Metering (NEM) scheme have been instrumental in expanding solar implementation across utility-scale, commercial, and residential sectors [13]. Despite this progress, integration of intermittent RE sources into the grid remains challenging due to variability in solar irradiation and feedstock supply for biomass. This limitation highlights the importance of energy storage deployment. As outlined in the Low Carbon Nation Aspiration 2040, Malaysia plans to expand the deployment of large-scale ESS during the Fourteenth and Fifteenth Malaysia Plans (2031–2040). Table 1 lists the profiles of selected solar energy companies in Malaysia that involve in energy storage development.

From a socio-environmental perspective, RE deployment offers several co-benefits. RE deployment can significantly reduce carbon footprints because RE generation produces far fewer lifecycle greenhouse gas (GHG) emissions than fossil-based generation [14]. RE also reduces reliance on fossil-based energy resources and supports broader decarbonisation objectives [15]. Moreover, RE adoption supports broader societal objectives, particularly by advancing the United Nations Sustainable Development Goals (SDGs), including Goal 7 (Affordable and Clean Energy) and Goal 13 (Climate Action).

Table 1. Profiles of selected solar energy companies in Malaysia

Company	Description	Source
Solarvest EE Sdn Bhd	A subsidiary of Solarvest Holdings Berhad, providing solar PV engineering, EPCC solutions in Malaysia.	https://solarvest.com/products-services/
Matrix Energy (M) Sdn Bhd	A Malaysian RE company involved in solar power development, offering EPC and project management services for commercial and LSS projects.	Information not publicly available
Sunsource Energy Sdn Bhd	A clean energy solutions provider in Malaysia, focusing on solar PV system design, installation, and maintenance for residential, commercial, and industrial clients.	https://www.sunsource.com.my

Note: PV = photovoltaic; EPCC = engineering, procurement, construction, and commissioning; RE = renewable energy; LSS = large-scale solar; EPC = engineering, procurement, and construction.

2.3 Energy Storage in Malaysia: Landscape and Policy

The deployment of energy storage in Malaysia is still nascent, marked by a small number of demonstration projects and pilot-scale initiatives across grid-scale systems, commercial and industrial (C&I) applications, residential users, microgrids, and island electrification. Early progress has been spearheaded by government-linked bodies such as Tenaga Nasional Berhad (TNB) and the Sustainable Energy Development Authority (SEDA), supported through frameworks outlined in the MyRER [1] and NETR [2]. These pilots are intended to test both the technical feasibility and economic value of BESS in the Malaysian context.

At present, lithium-ion batteries, especially those using LFP chemistries, are the foremost technology, with lead–acid systems still deployed in rural and backup settings due to their lower upfront cost, albeit with limited lifespans. Regional distribution of projects is concentrated in Peninsular Malaysia, while islanded systems, such as those in Pulau Tuba and Tioman, use storage to complement PV generation and reduce dependence on diesel. A landmark project is Malaysia's first sodium–sulfur (NaS) BESS (1.45 MWh), commissioned at the PV Large Scale Solar Farm in Bukit Selambau, Kedah. Although integrated with a solar facility, the system operates in grid support mode, demonstrating the potential role of future standalone BESS in load shifting, peak shaving, grid stability, and renewable firming [1].

Policy support for energy storage remains indirect. While Malaysia does not yet have a dedicated ESS regulatory framework, storage is increasingly acknowledged as a critical enabler of higher RE penetration. Current support is channelled through financial incentives such as the Green Investment Tax Allowance (GITA) and the Green Income Tax Exemption (GITE), which can apply when storage is bundled with renewable projects. However, the absence of explicit guidelines for interconnection, dispatch rights, cost recovery, and ancillary service markets limits large-scale deployment. Looking ahead, Malaysia has outlined plans to install five units of BESS with 100 MW capacity each year from 2030 to 2034 [16]. These initiatives mark an important step toward mainstreaming storage, though further policy clarity and market reforms will be essential to accelerate adoption and position Malaysia alongside regional leaders in the sector.

2.4 Energy Storage Systems: Classification and Technologies

ESS are increasingly recognised as indispensable enablers of RE integration, allowing surplus electricity generated during peak production to be stored and later discharged during periods of high demand. This functionality enhances overall grid reliability, stability, and operational flexibility [17]. Globally, a variety of storage technologies have been developed and deployed, each with distinct advantages and constraints. Electrochemical systems, particularly lithium-ion batteries such as LFP and NMC, dominate current markets due to their high energy density, rapid response, and sharply declining costs. Other electrochemical options, including lead–acid and flow batteries, remain relevant in specific contexts where cost or long-duration performance is prioritised [3]. Mechanical storage technologies, most notably pumped hydro storage, provide large-scale and long-duration capacity but are restricted by geographical and topographical requirements; alternatives such as compressed air and flywheels are being explored for niche applications. Thermal storage, which includes ice-based systems and molten salt, offers additional opportunities for balancing demand in district cooling and concentrated solar power applications. Finally, hydrogen-based storage is gaining prominence as a flexible, long-duration option that can couple renewable generation with industrial processes, presenting a potential pathway for future large-scale decarbonisation [18].

2.5 Performance Indicators of Energy Storage Systems

The performance of ESS is commonly assessed through a set of technical, economic, and operational indicators that capture their effectiveness in supporting RE integration and grid stability. Globally, the most widely cited technical indicators include RTE, which measures the ratio of energy discharged to energy charged, cycle life and degradation rate, which reflect the durability of storage technologies, and response time, which indicates suitability for fast frequency regulation [3, 10]. Economic indicators such as the LCOS, capital expenditure (CAPEX), and operating expenditure (OPEX) are central to evaluating the financial competitiveness of different technologies across use cases [5]. On the operational side, measures such as availability, reliability, and effective capacity contribution are important for quantifying the role of ESS in load shifting, peak shaving, and ancillary service provision [9]. Environmental indicators, particularly lifecycle greenhouse-gas emissions, are also relevant to holistic energy-system evaluation [14]. In contrast, the present review quantitatively compares how frequently these indicators have actually been reported across Malaysia-related ESS studies.

In the Malaysian context, however, existing studies and pilot projects provide only partial coverage of these performance indicators. While lithium-ion systems deployed in solar-linked microgrids report efficiency and reliability data, comprehensive evaluations of LCOS, degradation, or long-term environmental impacts remain limited. Furthermore, policy frameworks such as the MyRER and the NETR acknowledge ESS as a strategic enabler but do not prescribe standardised performance benchmarks. This gap highlights the need for a comparative framework of performance indicators tailored to Malaysia's energy transition, enabling both policymakers and industry actors to make evidence-based decisions on technology adoption and investment.

2.6 Performance Evaluation Framework for Energy Storage Systems

The preceding review demonstrates that research on ESS has expanded considerably in response to the increasing deployment of RE and the transition towards low-carbon power systems. Previous studies have examined a wide range of topics, including global developments in energy storage, RE deployment in Malaysia, national policy initiatives, storage technologies, and performance indicators. Collectively, these studies provide substantial evidence that ESS plays an important role in improving RE integration, enhancing grid flexibility, and supporting energy security. At the same time, the literature reflects the growing attention given to ESS within Malaysia's energy transition agenda through policy development, pilot projects, and increasing research activities.

Although the existing body of literature provides valuable insights into individual aspects of ESS, it remains dispersed across different research perspectives. Some studies focus on technology characteristics, others emphasise techno-economic feasibility, while several examine policy or RE integration. As a result, the available evidence is difficult to compare systematically because performance is evaluated using different objectives, application contexts, and analytical approaches. This fragmentation limits the ability of researchers, policymakers, and industry stakeholders to identify a consistent basis for assessing alternative storage technologies within Malaysia's evolving energy landscape.

These observations indicate the need for a more structured approach to evaluating ESS. Rather than considering technical, economic, operational, or policy-related aspects independently, a comprehensive evaluation should integrate these dimensions within a common decision-support framework. Similar multi-criteria engineering decision-making approaches have recently been applied to RE planning by integrating economic, technical, environmental, and implementation criteria, demonstrating the value of structured evaluation frameworks for supporting transparent technology selection and investment decisions [19].

To establish this framework, the present study adopts a systematic review based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 methodology. The reviewed evidence is synthesised to identify, compare, and organise the performance indicators reported in previous studies into a structured evaluation framework that is relevant to Malaysia's energy transition. The methodology employed to achieve this objective is described in the following section.

3 Method

3.1 Research Design

This study implements a systematic review methodology guided by the PRISMA 2020 framework [20], with additional elements of narrative synthesis to integrate findings across technical, economic, policy, and environmental dimensions. The primary aim is to identify and compare performance indicators of ESS relevant for Malaysia's energy transition.

To improve the transparency and traceability of the evidence synthesis, all retained studies were re-screened during the revision stage. This secondary eligibility assessment ensured that each retained study either directly contributed to the evaluation of ESS performance indicators or provided substantive contextual evidence relevant to their interpretation within the scope of Malaysia's energy transition.

3.2 Data Sources and Search Strategy

Literature was retrieved from four major academic databases: Scopus ($n = 64$), Web of Science ($n = 25$), IEEE Xplore ($n = 38$), and ScienceDirect ($n = 33$), resulting in a total of 160 records before screening. Official reports from the Energy Commission of Malaysia, SEDA, TNB, International Renewable Energy Agency (IRENA), and International Energy Agency (IEA) were used to support the background discussion and interpretation of findings but were not included in the PRISMA screening process. The search covered publications from 2000 to 2025 to capture both early developments and the most recent advances in ESS performance evaluation. The search string was refined to focus on performance-related indicators, as follows:

("energy storage" OR "battery energy storage system" OR "pumped hydro" OR "flow battery" OR "hydrogen storage") AND

("performance" OR "efficiency" OR "round-trip efficiency" OR "LCOS" OR "lifecycle cost" OR "reliability" OR "degradation") AND

("Malaysia" OR "Association of Southeast Asian Nations (ASEAN)" OR "Southeast Asia" OR "tropical climate")

3.3 Inclusion and Exclusion Criteria

Table 2 outlines the predefined inclusion and exclusion criteria adopted in this review. These criteria ensured a transparent and consistent selection process. Studies directly evaluating ESS performance indicators were retained as the core evidence base, whereas a limited number of closely related studies were retained as contextual evidence only when they addressed energy-system planning, RE integration, ESS policy, or transition pathways with substantive relevance to power-system ESS deployment. Studies focusing solely on electric vehicle (EV) battery management or portable battery/charging product development without power-system ESS evaluation were excluded in accordance with the predefined eligibility criteria. Contextual studies contributed only to the qualitative interpretation and were excluded from the quantitative indicator-frequency analysis. Hybrid hydrogen-based studies were eligible when battery and/or hydrogen storage constituted an explicit energy-storage component of the system and the study evaluated extractable technical, economic, or operational ESS performance indicators. Studies focusing solely on hydrogen production or fuel-cell performance without substantive evaluation of an energy-storage component were excluded.

Table 2. Inclusion and exclusion criteria

Criterion		Description
Inclusion	Study scope	Studies that directly evaluate, measure, model, compare, optimise, or systematically review ESS performance indicators.
	Technology	Battery, hydrogen, gravity, thermal, pumped hydro, compressed air, flywheel and other ESS technologies.
	Indicators	Technical, economic, operational, environmental or asset-management performance indicators.
	Context	Malaysia or directly relevant to Malaysia's energy transition.
	RE only	Studies focusing only on PV, wind or renewable generation without ESS performance evaluation.
Exclusion	Battery materials	Materials science, chemistry and electrode development without ESS performance evaluation.
	EV battery management	Studies focusing solely on EV battery management systems without evaluating ESS for power systems.
	Product development	Portable battery products or charging devices without ESS performance evaluation.
	Incidental ESS mention	Studies mentioning energy storage only incidentally, without ESS performance evaluation or substantive contextual relevance to power-system ESS deployment.
	Language	Non-English publications

Note: ESS = energy storage system; RE = renewable energy; PV = photovoltaic; EV = electric vehicle.

During the revision process, all initially included studies were re-evaluated against the predefined eligibility criteria. Studies with no substantive relevance to ESS performance or its interpretation were excluded from the final synthesis. Studies directly evaluating ESS performance indicators were retained as the core evidence base, while a limited number of closely related studies were retained only as contextual evidence where they supported the

interpretation of ESS development and performance. These contextual studies were excluded from all quantitative indicator-frequency calculations.

3.4 Screening and Selection Process

The screening and selection process followed the PRISMA 2020 guidelines to ensure transparency and reproducibility. Duplicate records were checked prior to screening, and no duplicate records were identified. The identified records were screened based on their titles and abstracts against the predefined inclusion and exclusion criteria. During this stage, 111 records were excluded because they focused primarily on RE technologies without ESS performance evaluation, laboratory-scale battery and material research, thermal-energy studies and building heating, ventilation, and air conditioning (HVAC) studies that did not evaluate electrical energy-storage performance within the predefined system-level scope, building energy management, HVAC applications, hydrogen production, fuel cells, EVs, or publications outside the predefined review scope.

Following full-text evaluation, nine studies were excluded because they did not satisfy the predefined inclusion criteria. Consequently, 40 studies were retained in the systematic review. Of these, 34 studies reported at least one explicitly extractable ESS performance indicator and constituted the core evidence base for the quantitative indicator-frequency analysis, while six studies were retained as contextual evidence and contributed only to the narrative synthesis. The study identification, screening, eligibility assessment, and final inclusion process are summarised in Figure 3.

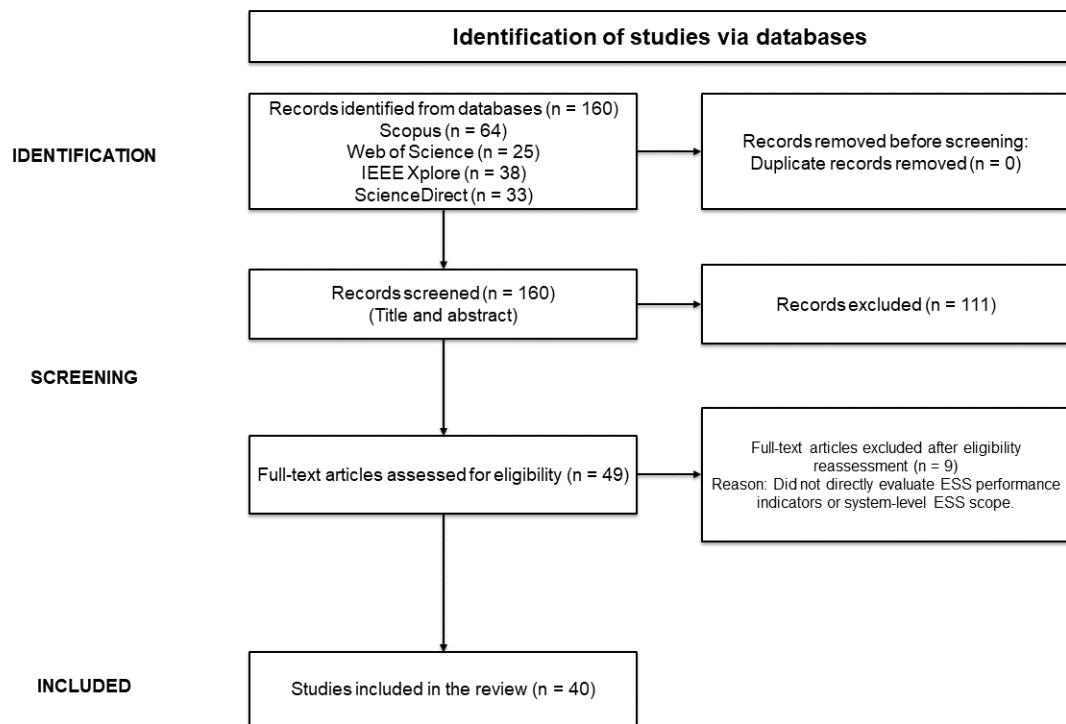


Figure 3. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 flow diagram of the study selection process

3.5 Data Extraction

For each included study, data were systematically extracted using a structured study-by-indicator matrix. The extracted performance indicators were organised into four dimensions:

- Technical indicators: capacity/sizing, general efficiency, RTE, degradation, lifetime/cycle life, response time, and state of charge/state of health/depth of discharge (SOC/SOH/DoD).
- Economic indicators: LCOS, levelized cost of energy (LCOE), net present cost (NPC)/lifecycle cost, CAPEX/OPEX/general cost analysis, and net present value (NPV)/internal rate of return (IRR)/payback period.
- Operational indicators: reliability/availability, ancillary services, peak shaving/demand reduction, load shifting/energy arbitrage, grid stability, voltage regulation, frequency regulation, and renewable integration/firming.
- Policy/environmental indicators: emissions/lifecycle GHG, recycling/end-of-life management, and safety/thermal management.

Each indicator was coded as E when it was explicitly evaluated, measured, modelled, reported, or systematically synthesised in the study; Q when it was discussed only qualitatively or contextually; and NR when it was not reported. The study-level classification “Contextual only” is distinct from the indicator-level code NR (“not reported”). A study was classified as “Contextual only” when it did not provide an E-coded performance indicator within the predefined system-level power-system ESS scope of this review. Such studies may nevertheless report measurable indicators relating to broader energy-transition, renewable-energy, policy, planning, or other system contexts; these measures were not counted as ESS performance indicators when they fell outside the predefined extraction framework. In contrast, NR indicates that a specific predefined ESS performance indicator was not reported in the study. The complete study-by-indicator coding is reported in Supplementary Table S1, which served as the authoritative extraction matrix for the quantitative analysis. Only entries coded E in this final matrix were included in the quantitative indicator-frequency analysis.

3.6 Analytical Framework

The extracted data were analysed using a comparative framework comprising four performance dimensions:

- Technical: capacity/sizing, efficiency, RTE, degradation, lifetime/cycle life, response time, and SOC/SOH/DoD.
- Economic: LCOS, LCOE, CAPEX/OPEX, NPC/lifecycle cost, and NPV/IRR/payback period.
- Operational: reliability/availability, grid stability, renewable integration, peak shaving, load shifting, voltage regulation, frequency regulation, and ancillary services.
- Policy/Environmental: emissions, safety/thermal management, and recycling/end-of-life management.

The framework enabled a consistent comparison of electrical energy storage technologies, including lithium-ion batteries, BESS, PHS, hydrogen energy storage, flow batteries, gravity energy storage, and hybrid ESS based on the extracted performance indicators.

Table 3. Engineering management coding framework

Code	Engineering Management Category	Decision Rule
EP	Engineering Planning	Assigned when the study focused on system planning, technology selection, storage sizing, capacity planning, network planning, RE integration, or infrastructure planning.
PE	Project Evaluation	Assigned when the study assessed techno-economic feasibility, project performance, lifecycle cost, LCOE, NPC, feasibility analysis, or comparative project evaluation.
ID	Investment Decision-making	Assigned when the study evaluated investment viability through CAPEX, OPEX, NPV, IRR, payback period, or economic decision support.
SO	System Operation	Assigned when the study investigated charging/discharging control, dispatch strategy, SOC/SOH management, EMS, grid operation, peak shaving, voltage regulation, or operational optimisation.
AM	Asset Management	Assigned when the study focused on degradation, battery lifetime, maintenance, reliability, remaining useful life, thermal management, or lifecycle extension.
SP	Strategic Planning	Assigned when the study addressed national policies, long-term energy transition, technology roadmaps, regulatory planning, institutional strategies, or national ESS deployment.

Note: RE = renewable energy; EMS = energy management system; ESS = energy storage system; LCOE = levelized cost of energy; NPC = net present cost; CAPEX = capital expenditure; OPEX = operating expenditure; NPV = net present value; IRR = internal rate of return; SOC = state of charge; SOH = state of health.

3.7 Engineering Management Coding Framework

To systematically analyse the engineering management perspective of the reviewed studies, each included study was classified according to its primary engineering management contribution. The classification was conducted using a predefined coding framework developed specifically for this review. Each study was assigned to one or more engineering management categories based on its stated research objectives, methodology, and reported outcomes.

Multiple classifications were permitted where a study addressed more than one engineering management function. To ensure consistency, the coding was performed using the decision rules presented in Table 3.

3.8 Quality and Relevance Appraisal

A structured quality and relevance appraisal was conducted for the 40 studies retained after application of the predefined eligibility criteria. Six criteria were assessed: (Q1) relevance to the review objective, (Q2) ESS technology relevance, (Q3) system-level ESS relevance, (Q4) performance-indicator evidence, (Q5) contribution to evidence synthesis, and (Q6) engineering-management relevance. Each criterion was scored on a three-point scale (2 = direct or explicit evidence, 1 = contextual or partial evidence, and 0 = no identifiable evidence), giving a maximum score of 12. The appraisal was used to characterise the relevance and evidentiary contribution of the retained studies and was not applied as an additional exclusion threshold. The same relevance-based appraisal criteria were applied consistently across the retained source types; source type itself was not used as a scoring criterion, and each source was assessed according to its actual evidentiary contribution to the review.

The appraisal was aligned with the study-by-indicator extraction in Supplementary Table S1 and the engineering-management coding framework applied in Table 4. Studies containing at least one explicitly reported performance indicator within the predefined system-level power-system ESS scope were classified as contributing to the quantitative indicator analysis. Studies that did not provide an E-coded indicator within this predefined scope, but provided substantive evidence relevant to ESS deployment, energy-system planning, policy, or transition pathways, were classified as “Contextual only” and retained for narrative synthesis. This classification does not imply that such studies contained no measurable indicators; rather, any measurable indicators they reported fell outside the predefined ESS performance-indicator framework used for quantitative frequency analysis. Accordingly, 34 of the 40 retained studies contributed explicit indicator evidence to the quantitative analysis, while six contributed contextual evidence to the narrative synthesis. The complete appraisal criteria, scoring rules, and study-level results are provided in Supplementary Table S2.

Table 4. Screened articles included in the review

No.	Study Focus	Technology	Application	Performance Indicators Extracted	Engineering Management Relevance	Indicator Extraction	Year	Ref.
1	Techno-economic evaluation of BESS	BESS	Corporate energy sharing	Capacity/sizing; energy sharing; CAPEX; OPEX; NPV; carbon-emission reduction	EP, PE, ID	Yes	2025	[21]
2	Power-system planning and optimisation	Multiple ESS technologies	Utility-scale power system	Capacity/sizing; power capacity; charging and discharging efficiency; electricity stored and discharged; CEEP reduction; import reduction; supply–demand balance	EP, SO	Yes	2025	[22]
3	Bibliometric assessment of energy research	Multiple energy technologies	Malaysian energy-research landscape	Publication trends; technology trends; research themes; collaboration patterns	SP	Contextual only	2025	[23]
4	Techno-economic evaluation of hybrid ESS	Hybrid ESS	Campus energy system	Capacity/sizing; system sizing; NPC; LCOE; renewable fraction; project feasibility	EP, PE, ID	Yes	2025	[24]
5	National electricity-transition modelling	BESS and pumped-hydro storage	National power system	Capacity/sizing; power generation; renewable integration; dispatch; electricity imports; carbon emissions; system cost	EP, SP	Yes	2025	[25]
6	BESS control and peak-demand reduction	Lithium-ion BESS	Peak-demand management	SOC; charging and discharging control; peak-demand reduction; forecasting accuracy; operational response	SO, AM	Yes	2025	[26]
7	BESS-based demand-side management	BESS	Urban distribution network	Capacity/sizing; peak shaving; load shifting; network losses; voltage performance; demand reduction	EP, SO	Yes	2024	[27]
8	Energy-transition review	Multiple energy and storage technologies	Malaysian energy transition	Technology readiness; policy support; transition barriers; planning requirements	EP, SP	Contextual only	2024	[28]

9	Rule-based BESS dispatch control	BESS	PV integration	SOC; charging and discharging schedule; dispatch strategy; battery operating condition; renewable-energy utilisation	SO	Yes	2024	[29]
10	Techno-economic evaluation of hydrogen storage	Hydrogen energy storage	EV charging station	Hydrogen-capacity/sizing; electrolyser and fuel-cell sizing; efficiency; LCOE; NPC; energy balance	EP, PE, ID	Yes	2024	[30]
11	National energy-transition assessment	Multiple energy technologies	National power system	Renewable-energy contribution; energy security; transition pathways; policy and planning indicators	EP, SP	Contextual only	2024	[31]
12	ESS benefits and application review	Multiple ESS technologies	Malaysian power system	Power capacity; capacity/sizing; charging and discharging characteristics; grid stability; voltage support; peak shaving; ancillary services; renewable integration	EP, SP	Yes	2024	[32]
13	BESS planning and microgrid design	BESS	Net Energy Metering	Battery capacity; system sizing; energy balance; self-consumption; electricity-cost reduction; project feasibility	EP, PE	Yes	2024	[33]
14	BESS deployment and grid-integration review	Lithium-ion BESS	Malaysian electricity grid	Energy and power capacity; SOC; efficiency; reliability; voltage regulation; frequency regulation; active and reactive power; power quality; grid stability; degradation; operational lifespan	EP, ID, SO	Yes	2024	[16]
15	Carbon-neutrality pathway assessment	Multiple energy technologies	National decarbonisation	Renewable-energy contribution; carbon-emission reduction; energy-system transition; policy readiness	EP, SP	Contextual only	2024	[34]
16	Residential solar-PV system review	Residential PV with battery storage	Residential PV systems	Battery integration; self-consumption; backup supply; intermittency management; system reliability; smart-grid integration	EP, SP	Contextual only	2023	[35]
17	ASEAN power-sector decarbonisation assessment	Multiple energy technologies	ASEAN power system	Renewable penetration; emissions reduction; system transition; regional policy and planning	EP, SP	Contextual only	2023	[36]
18	ESS progress and distribution-network review	Multiple ESS technologies	Malaysian distribution network	Power density; energy density; discharge duration; efficiency; response time; lifespan; cycle life; degradation; remaining capacity; peak shaving; load shifting; arbitrage; operating reserve; ancillary services; capital cost; distribution deferral	EP, SO, SP	Yes	2023	[37]
19	Optimal sizing of battery–hydrogen storage	Battery–hydrogen hybrid ESS	Renewable power system	Battery and hydrogen capacity; degradation; lifecycle cost; NPC; LCOE; reliability; seasonal-storage performance	EP, PE, ID, AM	Yes	2023	[38]
20	Techno-economic feasibility assessment	Battery storage	Conservation-park microgrid	Battery capacity; autonomy; renewable fraction; NPC; LCOE; payback period; emissions reduction	EP, PE, ID	Yes	2023	[39]
21	Industrial hybrid-energy-system modelling	Battery storage	Industrial demand management	Battery capacity; peak shaving; load shifting; demand reduction; energy cost; operational scheduling	EP, PE, SO	Yes	2023	[40]
22	Utility-scale BESS integration	BESS	Utility-scale solar PV	BESS capacity; power rating; renewable curtailment reduction; grid support; peak reduction; techno-economic performance	EP, PE, SO	Yes	2023	[41]
23	PV–battery system sizing	BESS	Commercial building	Battery capacity; PV and storage sizing; energy balance; NPC; LCOE; payback period; self-consumption	EP, PE, ID	Yes	2023	[42]

24	Techno-economic-environmental evaluation	Hybrid ESS	Renewable power system	Capacity/sizing; system efficiency; LCOE; NPC; emissions; environmental impact; reliability	PE, ID, AM	Yes	2022	[43]
25	ASEAN solar-energy, energy-storage and policy review	Multiple energy and storage technologies	ASEAN renewable-energy and storage systems	Storage capacity; LCOE; NPC; cost/investment; NPV/payback; reliability; grid stability; voltage support; frequency regulation; SOC; emissions	PE, ID, SO, SP	Yes	2022	[44]
26	Gravity-storage feasibility assessment	Gravity energy storage	Residential energy system	Capacity/sizing; power rating; efficiency; depth of discharge; cycle life; capital cost; operating cost; carbon footprint	EP, PE	Yes	2022	[45]
27	Battery energy-management system	Battery storage	Rural electrification	SOC; charging and discharging control; energy balance; battery capacity; load management; supply reliability	EP, SO	Yes	2022	[46]
28	Gravity-storage suitability assessment	Gravity energy storage	Malaysian power system	Capacity/sizing; efficiency; cycle life; response characteristics; cost; grid suitability; environmental performance	EP, PE	Yes	2022	[47]
29	PV–diesel–battery performance evaluation	Battery storage	Rural electrification	Battery capacity; NPC; LCOE; renewable fraction; unmet load; fuel consumption; emissions; reliability	PE, SO	Yes	2021	[48]
30	Profitability assessment of PV–battery systems	Battery storage	C&I sector	Battery capacity; RTE; degradation; lifetime/cycle life; SOC/DoD; CAPEX/OPEX; electricity savings; NPV; IRR; peak-demand reduction; energy arbitrage	PE, ID	Yes	2020	[49]
31	Battery power-pack modelling	Battery storage	Island electrification	Battery voltage; charging and discharging behaviour; discharge characteristics; capacity/sizing; operational performance	EP, SO	Yes	2019	[50]
32	Battery-lifetime enhancement	Battery–supercapacitor hybrid ESS	Standalone PV system	Battery lifetime; degradation; SOC; depth of discharge; charge–discharge stress; power sharing; lifecycle extension	SO, AM	Yes	2019	[51]
33	BESS sizing optimisation	BESS	Low-voltage distribution network	Battery capacity; SOC; peak-demand reduction; reverse power flow; voltage performance; storage sizing	EP, PE	Yes	2018	[52]
34	Smart-energy-system modelling	Multiple ESS technologies	Tropical energy system	Capacity/sizing; charging and discharging schedule; renewable integration; curtailment; supply–demand balance; system optimisation	EP, SP	Yes	2018	[53]
35	Economic sizing of hydrogen storage	Hydrogen energy storage	Standalone PV system	Hydrogen Capacity/sizing; electrolyser sizing; fuel-cell sizing; system efficiency; lifecycle cost; LCOE	EP, PE, ID	Yes	2017	[54]
36	Wind–PV–battery system integration	Battery storage	Rural electrification	Battery capacity; SOC; renewable fraction; unmet demand; system reliability; energy balance	EP, SO	Yes	2017	[55]
37	ESS-based voltage regulation	ESS	Distribution network	Voltage deviation; voltage regulation; active and reactive power support; grid stability; control response	SO	Yes	2017	[56]
38	PV–diesel–battery performance evaluation	Battery storage	Rural electrification	Battery capacity; NPC; LCOE; renewable fraction; unmet load; fuel consumption; emissions; reliability	PE, SO	Yes	2017	[57]
39	Smart battery power management	Battery storage	Microgrid	SOC; battery dispatch; charging and discharging control; load sharing; energy management; operational reliability	SO, AM	Yes	2014	[58]

40	Hybrid renewable system performance	Battery and hydrogen storage	Hydrogen production	Battery capacity; hydrogen production; hydrogen-storage capacity; system efficiency; renewable contribution; energy balance; cost performance	EP, PE, SO	Yes	2009	[59]
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Note: ESS = energy storage system; BESS = battery energy storage system; PV = photovoltaic; CAPEX = capital expenditure; OPEX = operating expenditure; NPV = net present value; CEEP = critical excess electricity production; NPC = net present cost; LCOE = levelized cost of energy; SOC = state of charge; RTE = round-trip efficiency; DoD = depth of discharge; IRR = internal rate of return; EP = engineering planning; PE = project evaluation; ID = investment decision-making; SO = system operation; AM = asset management; SP = strategic planning; EV = electric vehicle; ASEAN = Association of Southeast Asian Nations; C&I = commercial and industrial.

3.9 Limitations

The review is constrained by the availability of public data, particularly on commercial BESS projects where operational and financial details are often proprietary. In such cases, grey literature and triangulation across multiple sources were used to mitigate gaps.

4 Results and Discussion

4.1 Descriptive and Bibliometric Profile of the Included Studies

This section presents the findings of the systematic literature review based on the 40 studies retained after the PRISMA-guided screening and eligibility verification process. A descriptive analysis was first performed to characterise the included studies in terms of publication trends, document types, source titles, authorship, technologies, and application domains (Figure 4, Figure 5, and Table 4). Subsequently, the quantitative indicator-frequency analysis focused on the studies that explicitly reported extractable ESS performance indicators, whereas the remaining studies contributed to the narrative synthesis.

Table 4 provides an abbreviated descriptive summary of the retained studies and selected extracted indicators; it is not intended to reproduce the complete study-by-indicator coding. The complete extraction matrix, including the E (explicit), Q (qualitative/contextual), and NR (not reported) coding for each performance indicator and each retained study, is provided in Supplementary Table S1. Supplementary Table S1 constitutes the authoritative extraction matrix used to derive and validate all indicator-frequency counts reported in Figure 6, Figure 7, Figure 8, and Figure 9. Only indicators coded E were included in the quantitative frequency calculations. Studies labelled “Contextual only” did not contribute to these frequency counts. “Contextual only” denotes the study-level role in the synthesis and should not be interpreted as equivalent to NR; NR is an indicator-level code indicating that a specific predefined ESS performance indicator was not reported.

Figure 4 presents the distribution of publications by year, journal outlet, author productivity, and document type. These patterns provide insights into the evolution of energy storage research relevant to Malaysia and situate the present review within a rapidly expanding scholarly domain.

The demographic profile of the 40 included studies indicates a growing research interest in ESS relevant to Malaysia, particularly from 2022 onwards, with the highest publication output recorded in 2024. Journal articles constituted the dominant document type, followed by conference papers and review articles. The Journal of Energy Storage was the most frequent publication outlet, while contributions were also distributed across several high-impact journals and conference proceedings, reflecting the multidisciplinary nature of ESS research. Authorship analysis demonstrates active collaboration among both Malaysian and international researchers, highlighting the increasing research capacity and global engagement in this field. Overall, these publication trends indicate that ESS has become an increasingly important research area supporting Malaysia’s energy transition, providing a timely foundation for the subsequent comparative evaluation of ESS performance indicators.

Figure 5 presents the term co-occurrence network derived from the titles and abstracts of the included studies. Five interconnected thematic clusters can be observed. The red cluster is associated with modelling and feasibility assessment, represented by terms such as modelling, electricity, and solar energy. The green cluster reflects RE and distribution-network integration, linking renewable energy, PV system, and distribution network. The blue cluster is centred on battery energy storage control and effectiveness, with prominent terms including battery energy storage, control scheme, and effectiveness. The yellow cluster is primarily associated with operational and dispatch strategies, particularly dispatch strategy and its connections with feasibility, renewable-energy integration, and battery storage. The purple cluster represents the broader renewable-system context, linking RE source, hybrid renewable energy system (HRES), and viability.

The network demonstrates that the literature is strongly interconnected around renewable-energy integration, system modelling, dispatch strategies, and BESS control. In contrast, explicit performance indicators such as LCOS, degradation, and ancillary-service metrics do not emerge as prominent terms in the network. This observation complements the study-level performance-indicator analysis presented in the subsequent sections and further indicates

that the Malaysian ESS literature has concentrated more strongly on system integration and operational applications than on comprehensive multidimensional performance assessment.



Figure 4. Demographic profile of the screened literature ($n = 40$): (a) distribution of publications by year; (b) most active source titles; (c) most productive authors (all co-authors); and (d) distribution by document type

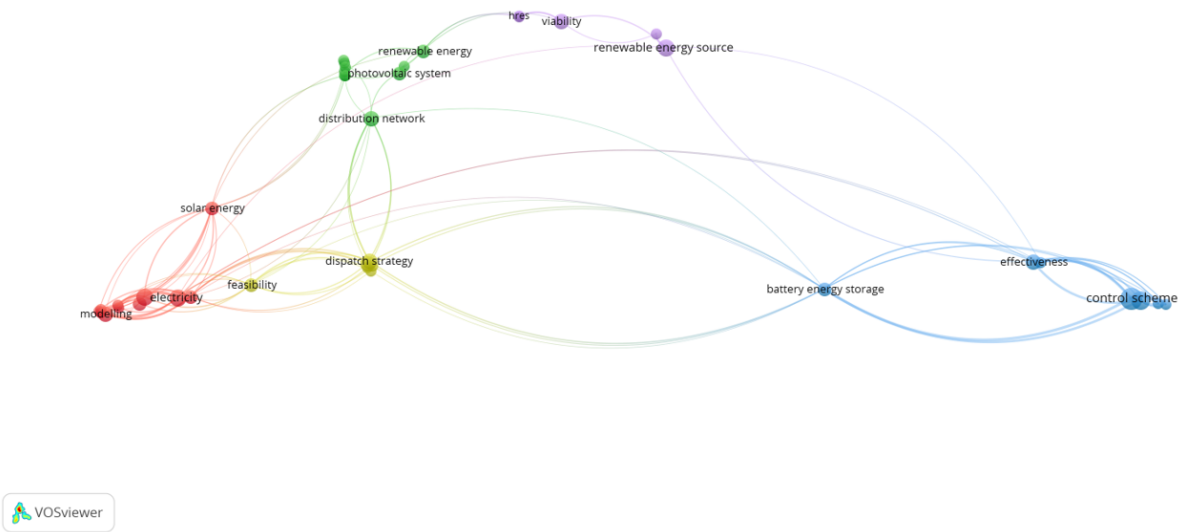


Figure 5. Term co-occurrence network based on titles and abstracts of the included studies

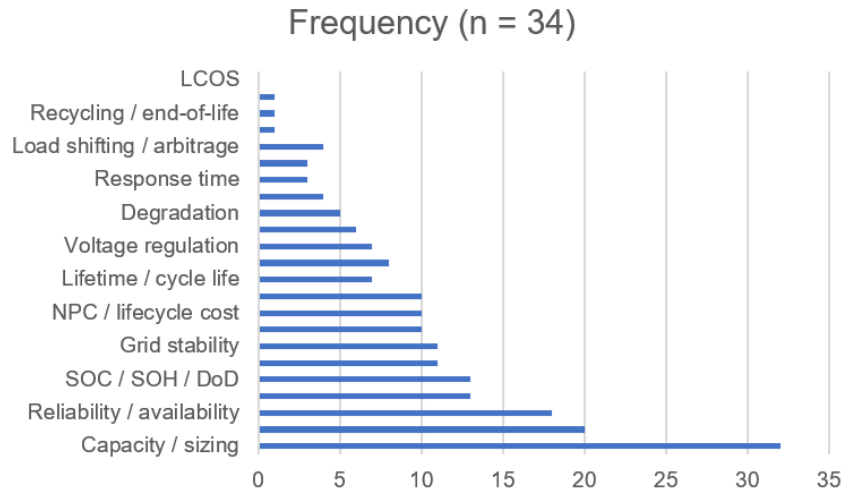


Figure 6. Frequency of ESS performance indicators extracted from the 34 studies reporting explicit ESS performance indicators

Note: ESS = energy storage system; SOC = state of charge; SOH = state of health; DoD = depth of discharge; NPC = net present cost; LCOS = levelized cost of storage.

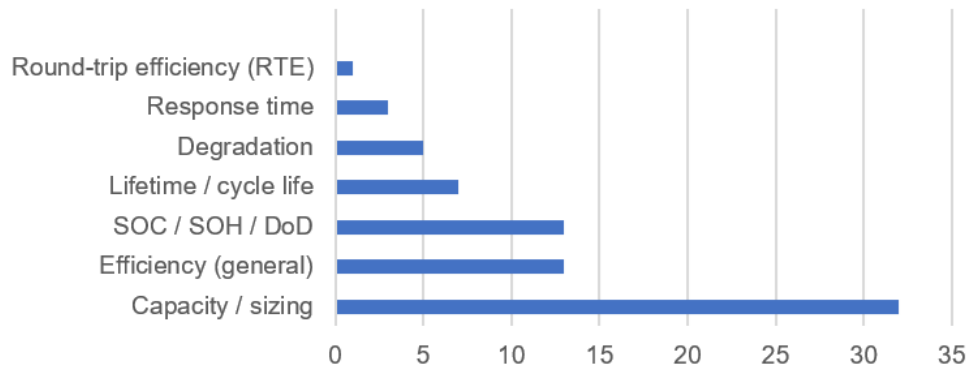


Figure 7. Frequency of technical performance indicators extracted from the 34 studies reporting explicit ESS performance indicators

Note: ESS = energy storage system; SOC = state of charge; SOH = state of health; DoD = depth of discharge.

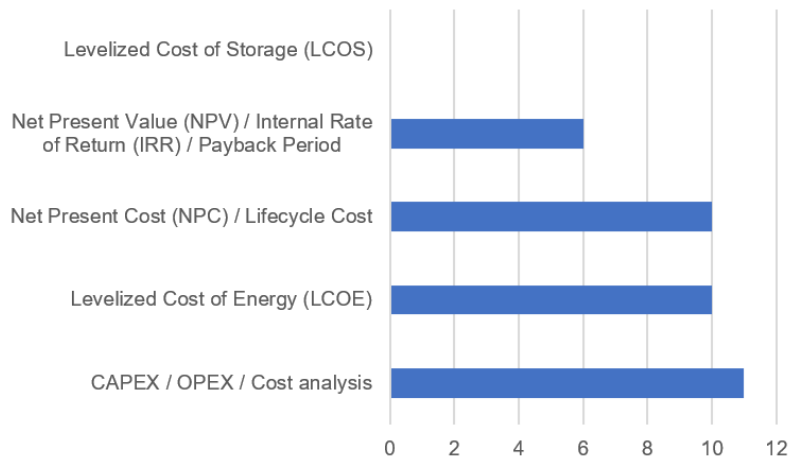


Figure 8. Frequency of economic performance indicators extracted from the 34 studies reporting explicit ESS performance indicators

Note: ESS = energy storage system; CAPEX = capital expenditure; OPEX = operating expenditure.

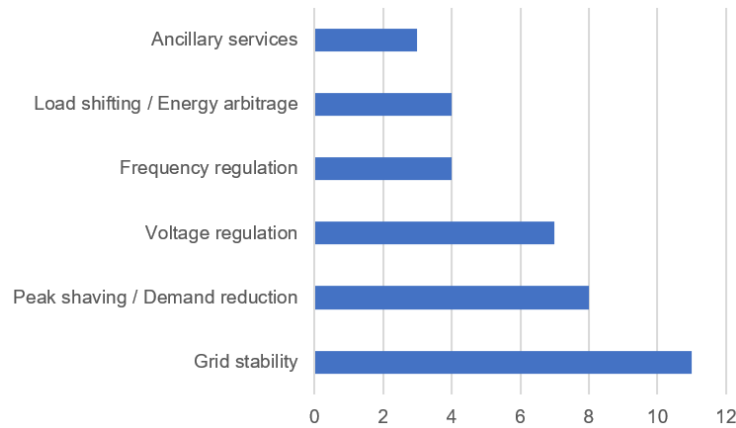


Figure 9. Frequency of operational and grid-related performance indicators extracted from the 34 studies reporting explicit energy storage system (ESS) performance indicators

4.2 Energy Storage Systems Performance-Indicator Frequency Analysis

Assessing the performance of ESS requires clear indicators that capture their technical efficiency, economic viability, operational reliability, and policy relevance. However, the literature addressing ESS in Malaysia and the wider ASEAN region often applies these indicators inconsistently, limiting comparability across studies. To provide a structured overview, the screened articles were analysed to identify the most frequently reported performance indicators, the technologies under investigation, and the application contexts in which storage is deployed. All frequencies reported in this section were derived from and cross-validated against the final study-by-indicator extraction matrix in Supplementary Table S1; only indicators coded as explicitly evaluated (E) were counted.

Figure 6 summarises the frequency of ESS performance indicators reported across the 34 studies contributing to the quantitative indicator analysis. Capacity and sizing emerged as the most frequently evaluated indicator ($n = 32$), reflecting the strong emphasis on storage sizing, system configuration, and planning for RE integration, particularly in PV-coupled, hybrid, and utility-scale applications, as demonstrated across BESS, hybrid ESS, hydrogen-storage, and grid-integration studies [21, 24, 27, 30, 33, 38, 41, 42, 52]. Renewable integration or firming ($n = 20$) and reliability or availability ($n = 18$) were the next most frequently reported indicators, indicating that much of the reviewed literature focuses on the role of ESS in supporting renewable-energy integration and reliable power-system operation [22, 24, 25, 32, 37–39, 46, 48, 53, 57, 58].

General efficiency ($n = 13$), SOC/SOH/DoD ($n = 13$), grid stability ($n = 11$), CAPEX/OPEX ($n = 11$), LCOE ($n = 10$), NPC/lifecycle cost ($n = 10$), and emissions/lifecycle GHG ($n = 10$) also received considerable attention, indicating that technical and conventional techno-economic assessments dominate existing ESS evaluation. By contrast, degradation was explicitly evaluated in only five studies ($n = 5$) [16, 37, 38, 49, 51], while frequency regulation was reported in four studies ($n = 4$) [16, 37, 44, 56]. Response time ($n = 3$) [26, 37, 56], ancillary services ($n = 3$) [16, 32, 37], and load shifting or energy arbitrage ($n = 4$) [27, 37, 40, 49] were also infrequently evaluated. RTE was explicitly evaluated in only one study [49], while recycling/end-of-life management [37] and safety/thermal management [16] were each explicitly addressed in only one study. No retained study explicitly evaluated LCOS.

Overall, the distribution suggests that the reviewed literature remains strongly oriented toward system planning, renewable integration, and project feasibility, while long-term operational performance, lifecycle management, and market-oriented ESS services receive comparatively limited attention. The absence of LCOS is particularly significant because it limits consistent economic comparison among storage technologies using a storage-specific cost metric.

Figure 7 presents the technical performance indicators reported across the reviewed studies. Capacity or sizing remained the most frequently evaluated technical indicator ($n = 32$), reflecting the strong emphasis on determining appropriate storage capacity, battery sizing, and system configuration for ESS deployment, particularly in PV-coupled systems, hybrid RE systems, and utility-scale applications [21, 22, 24, 25, 27, 30, 33, 38, 41, 42, 52]. This finding indicates that the reviewed literature primarily focuses on engineering design and system planning, where appropriate storage sizing is fundamental for achieving reliable system operation and effective energy management.

General efficiency ($n = 13$) was also frequently evaluated, particularly in studies assessing charging/discharging performance, hybrid system efficiency, and energy conversion [16, 22, 30, 32, 37, 43, 45, 47, 50, 51, 53, 54, 59]. SOC/SOH/DoD was reported in 13 studies, particularly those focusing on battery control, dispatch, energy management, and battery utilisation [16, 26, 29, 37, 44–46, 49–52, 55, 58].

Conversely, degradation was explicitly evaluated in only five studies [16, 37, 38, 49, 51], while response time

was evaluated in three studies [26, 37, 56]. Most notably, RTE was explicitly evaluated in only one study [49]. Although degradation and RTE are internationally recognised as important technical indicators for comparing ESS technologies, they remain underrepresented in the reviewed literature. This suggests that existing studies have primarily concentrated on system design and operational feasibility, while comparatively limited attention has been given to long-term battery behaviour, energy conversion efficiency, and dynamic performance evaluation.

Figure 8 summarises the economic performance indicators reported across the reviewed studies. CAPEX/OPEX or general cost analysis represented the most frequently evaluated economic indicator ($n = 11$) [21, 25, 33, 37, 40, 41, 44, 45, 47, 49, 59]. LCOE ($n = 10$) and NPC/lifecycle cost ($n = 10$) were also widely reported, with both indicators appearing in studies [24, 30, 38, 39, 42–44, 48, 54, 57]. Collectively, these findings indicate that the reviewed literature primarily emphasises conventional project-cost evaluation and economic feasibility during the planning stage of ESS deployment.

NPV, IRR, and payback period were reported in only six studies [21, 33, 39, 42, 44, 49], indicating that comprehensive investment appraisal remains relatively limited despite increasing interest in ESS deployment. More importantly, none of the reviewed studies explicitly reported the LCOS. This represents an important methodological gap because LCOS is internationally recognised as one of the most appropriate indicators for comparing alternative storage technologies using a common storage-specific economic metric. Its absence suggests that the reviewed literature continues to rely on conventional renewable-energy project economics rather than storage-specific lifecycle cost evaluation.

Figure 9 presents the frequency of operational performance indicators reported across the 34 studies contributing to the quantitative indicator analysis. Grid stability emerged as the most frequently reported operational indicator ($n = 11$), reflecting the important role of ESS in supporting secure grid operation, RE integration, operational flexibility, and overall system reliability [16, 22, 25, 32, 37, 56]. Peak shaving or demand reduction was evaluated in eight studies [26, 27, 32, 37, 40, 41, 49, 52], while voltage regulation was reported in seven studies [16, 27, 32, 37, 44, 52, 56]. Frequency regulation was evaluated in four studies [16, 37, 44, 56], while load shifting or energy arbitrage was reported in four studies [27, 37, 40, 49]. Ancillary services were explicitly evaluated in only three studies [16, 32, 37]. The reviewed literature suggests that ESS research has primarily focused on RE integration and demand management, whereas comparatively fewer studies have evaluated market-oriented operational services. From an engineering management perspective, this imbalance limits the evidence available to support investment planning, operational decision-making, and electricity market participation. Future studies should therefore evaluate these operational services using standardised performance metrics to strengthen project evaluation, asset management, and long-term ESS deployment strategies.

Overall, Figure 6, Figure 7, Figure 8, and Figure 9 demonstrate that current Malaysian ESS research is strongly concentrated on system sizing, renewable integration, technical feasibility, and conventional techno-economic evaluation. In contrast, internationally recognised performance indicators such as RTE, degradation, ancillary services, recycling, and safety remain substantially underreported, while LCOS was not explicitly evaluated in any of the retained studies.

Table 5 classifies the performance indicators into four complementary dimensions: technical, operational, economic, and policy/environmental. Although these dimensions are presented separately for analytical clarity, the reviewed literature suggests that they are conceptually interrelated in practical ESS evaluation. As illustrated in Figure 10, technical performance is conceptually linked to system operation and reliability, while both technical and operational performance may influence the economic evaluation of ESS projects. Policy and environmental conditions, including regulatory frameworks, market mechanisms, safety standards, and recycling requirements, may also shape how technical, operational, and economic performance is evaluated in different deployment contexts. Therefore, the four dimensions are interpreted in this review as an integrated systems-oriented framework rather than as independent categories, providing a structured basis for evaluating ESS within SSCM.

This conceptual interdependence is also reflected in the reviewed literature. Several techno-economic studies evaluated technical indicators such as efficiency together with CAPEX, OPEX, and project-feasibility measures, suggesting an association between technical and economic considerations in ESS evaluation. Similarly, studies on battery dispatch control and energy-management systems considered operational strategies alongside technical capabilities such as response time and capacity/sizing, while policy-oriented studies highlighted the relevance of regulatory support and market mechanisms to ESS deployment. These patterns do not establish causal relationships among the four dimensions; rather, they indicate that the dimensions are frequently considered together and may interact within ESS planning, evaluation, and management. Accordingly, Figure 10 represents a conceptual synthesis of the reviewed evidence rather than an empirically tested causal model.

Based on the integrated evaluation framework, Table 6 maps the 34 studies contributing explicit performance-indicator evidence across broad, mutually exclusive ESS technology groups and the four performance dimensions. For this cross-technology synthesis, the study-level technology descriptions in Table 4 were consolidated into five broad groups to avoid overlap between technology type, storage configuration, and application context. Each study

was assigned to one technology group only, and a study was counted within a performance dimension when at least one indicator in that dimension was coded E in Supplementary Table S1.

Table 5. Classification of ESS performance indicators

Technical Indicators	Economic Indicators	Operational Indicators	Policy/Environmental Indicators
RTE; efficiency (general); degradation/cycle life; response time; capacity/sizing; SOC/SOH/DoD	CAPEX/OPEX; LCOE/LCOS; NPC/lifecycle cost; NPV/IRR/payback	Availability/reliability; ancillary services/frequency regulation; peak shaving/demand charge management; load shifting/arbitrage; renewable integration/firming; curtailment reduction; grid stability; voltage regulation	Safety; recycling; emissions
Note: ESS = energy storage system; RTE = round-trip efficiency; SOC = state of charge; SOH = state of health; DoD = depth of discharge; CAPEX = capital expenditure; OPEX = operating expenditure; LCOE = levelized cost of energy; LCOS = levelized cost of storage; NPC = net present cost; NPV = net present value; IRR = internal rate of return.			

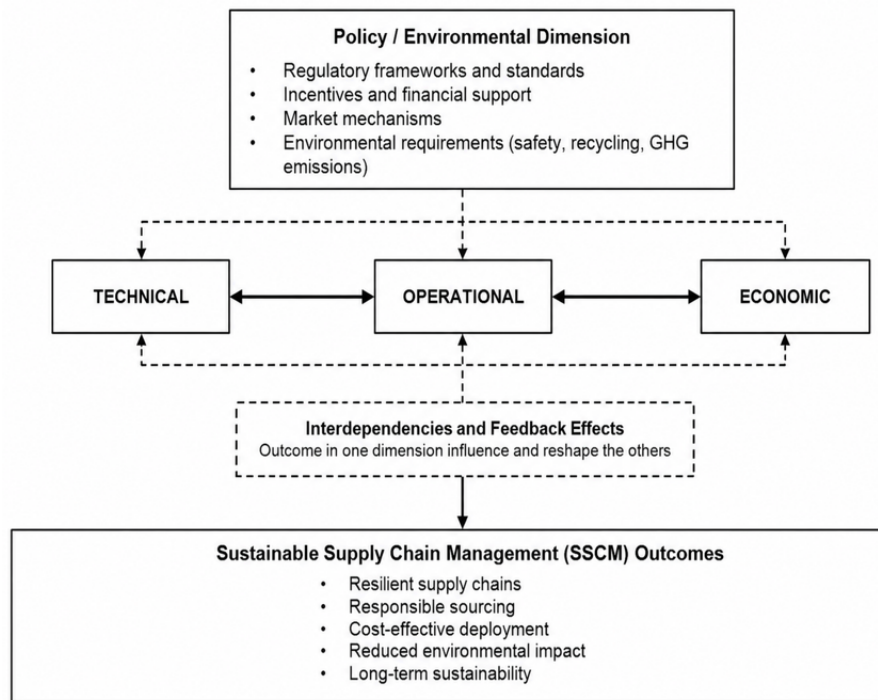


Figure 10. Conceptual framework for evaluating ESS performance

Note: ESS = energy storage system; GHG = greenhouse gas.

Table 6 shows that battery-based ESS constitute the largest technology group in the quantitative evidence base ($n = 18$), with all 18 studies reporting at least one technical and one operational performance indicator. Economic indicators were explicitly evaluated in nine battery-based studies, while policy/environmental indicators were represented in five. Hybrid and multi-technology ESS constituted the second-largest group ($n = 11$), with technical and operational indicators represented across all 11 studies, economic indicators in seven studies, and policy/environmental indicators in five. Hydrogen energy storage was represented by two studies, both covering technical, economic, and operational dimensions but neither explicitly evaluating the policy/environmental indicators defined in this review. Gravity energy storage was also represented by two studies, both addressing technical, economic, and policy/environmental indicators, while one addressed operational performance. The single generic ESS study contributed technical and operational evidence only.

Overall, the cross-technology synthesis reinforces the predominance of battery-based and hybrid/multi-technology ESS in the reviewed evidence. It also shows that technical and operational evaluation is more consistently represented across technology groups than policy/environmental assessment.

Table 6. ESS technology–performance dimension coverage

Technology	Technical	Economic	Operational	Policy/ Environment	Colour Scale
Battery-based ESS	18	9	18	5	13–18
Hybrid/multi-technology ESS	11	7	11	5	9–12
Hydrogen energy storage	2	2	2	0	5–8
Gravity energy storage	2	2	1	2	1–4
Generic ESS	1	0	1	0	0

Note: ESS = energy storage system; E = explicitly evaluated. Values represent the number of studies within each mutually exclusive technology group reporting at least one E-coded indicator in the corresponding performance dimension. Colour intensity represents study coverage among the 34 studies reporting explicit ESS performance indicators and does not represent the total number of extracted indicators.

4.3 Engineering Management Perspective of the Reviewed Studies

To strengthen the engineering management perspective, the 40 retained studies were further interpreted according to six engineering management functions, while the 34 studies reporting explicit ESS performance indicators constituted the core evidence base for the quantitative indicator analysis. The synthesis shows that studies on energy system modelling, technology selection, microgrid design, RE integration, and storage sizing primarily support engineering planning, enabling systematic selection and configuration of ESS according to technical requirements and application needs. Project evaluation and investment decision-making are predominantly informed by techno-economic, feasibility, and profitability studies integrating CAPEX/OPEX, LCOE, NPC/lifecycle cost, NPV, IRR, payback period, and environmental performance. However, the absence of explicit LCOS evaluation indicates that storage-specific lifecycle economic assessment remains a major gap in the reviewed evidence. Studies focusing on dispatch control, demand-side management, voltage regulation, and energy management systems (EMS) contribute to system operation, highlighting the importance of availability, reliability, response time, peak shaving, and load shifting for operational optimisation. Meanwhile, research on battery degradation, lifetime enhancement, battery management systems, safety, and recyclability supports lifecycle and asset management by informing maintenance planning, replacement strategies, and end-of-life management. Finally, policy reviews and national energy transition studies emphasise strategic policy planning, demonstrating how regulatory frameworks, market mechanisms, and sustainability requirements influence ESS deployment.

Table 7. Engineering management interpretation of the reviewed evidence

Engineering Management Function	Evidence Synthesised from the Reviewed Studies	Performance Indicators Involved	Engineering Management Decisions Supported
Engineering planning	Studies on system sizing, energy modelling, technology selection and renewable integration	Efficiency, RTE, capacity, response time	Technology selection, system configuration, planning
Project evaluation	Techno-economic, feasibility and environmental assessment studies	CAPEX/OPEX, LCOE, NPC/lifecycle cost, emissions	Project comparison and feasibility
Investment decision-making	Profitability, economic analysis, sensitivity studies	Payback, NPV, IRR	Investment prioritisation
System operation	Dispatch, peak shaving, voltage regulation, EMS	Reliability, availability, load shifting	Operational optimisation
Lifecycle & asset management	Battery degradation, BMS, lifetime studies	Cycle life, degradation, safety	Maintenance, replacement planning
Policy & strategic management	NETR, MyRER, policy reviews	Environmental, recycling, policy	Strategic planning

Note: RTE = round-trip efficiency; CAPEX = capital expenditure; OPEX = operating expenditure; LCOE = levelized cost of energy; NPC = net present cost; NPV = net present value; IRR = internal rate of return; EMS = energy management system; BMS = battery management system; NETR = National Energy Transition Roadmap; MyRER = Malaysia Renewable Energy Roadmap.

Overall, the reviewed evidence indicates that the proposed framework extends beyond performance assessment by providing a structured decision-support approach for engineering management. However, the literature remains concentrated on technical performance and preliminary techno-economic assessment, with comparatively limited

attention to LCOS, degradation, ancillary services, recycling, and long-term asset management. Addressing these gaps is essential to support more comprehensive engineering planning and sustainable ESS deployment in Malaysia. To better illustrate the engineering management contribution of the reviewed literature, the 40 retained studies were further synthesised based on their engineering management relevance. As shown in Table 7, the studies were classified into six engineering management functions according to their objectives, application context, and reported performance indicators.

Furthermore, Table 7 indicates that the reviewed studies cover multiple engineering management functions rather than focusing solely on technical performance. Most studies contribute to engineering planning, project evaluation, and system operation through energy system modelling, techno-economic assessment, and operational optimisation. In contrast, comparatively fewer studies address lifecycle asset management and long-term strategic planning, particularly those related to degradation, recycling, end-of-life management, and policy-driven decision-making. This finding supports the need for an integrated performance evaluation framework that incorporates technical, economic, operational, and policy/environmental dimensions to facilitate engineering management decisions throughout the ESS lifecycle.

The proposed framework can be applied to compare alternative energy storage technologies by integrating technical, operational, economic, and policy or environmental indicators into a unified evaluation. It also supports decision-making throughout the project lifecycle, from technology selection and feasibility assessment to operational management and asset replacement. Under different policy scenarios, the framework enables stakeholders to evaluate the implications of regulatory changes on technology selection, investment decisions, and sustainable supply chain planning.

5 Conclusions

This systematic review included 40 eligible studies. Of these, studies reporting explicitly extractable ESS performance indicators contributed to the quantitative indicator synthesis, while the remaining studies informed the narrative interpretation of policy, planning, and implementation issues. The findings demonstrate that, although ESS research has expanded considerably in recent years, the evaluation of performance indicators remains uneven. Capacity/sizing, renewable integration/firming, and reliability/availability are the most frequently reported performance indicators, while general efficiency and conventional techno-economic measures receive moderate attention. In contrast, RTE, degradation, ancillary services, and lifecycle sustainability remain underreported. No retained study explicitly reported LCOS. Battery-based ESS, particularly BESS, are the most widely investigated storage technologies, particularly in PV-coupled applications, while long-duration energy storage technologies, including flow batteries, pumped hydro, and hydrogen storage, remain underexplored. Compared with countries that have more mature ESS deployment, Malaysia is still at an early stage of adopting comprehensive performance evaluation practices, particularly for operational and market-based indicators.

These findings have important implications for policy and future research. The limited consideration of advanced performance indicators highlights the need to establish national ESS performance benchmarks to support consistent technology evaluation and comparison. The predominance of PV-coupled battery applications suggests that financial incentives should be expanded to encourage broader deployment of standalone ESS technologies. Furthermore, the limited assessment of ancillary services reflects the need for dedicated market mechanisms and clearer regulatory frameworks to facilitate commercial ESS adoption. Future research should prioritise field-based investigations of long-term operational indicators, including LCOS, degradation behaviour, lifecycle sustainability, and system-level performance under Malaysia's tropical operating conditions. Collectively, these recommendations are directly informed by the evidence synthesised in this review and provide practical guidance for strengthening SSCM and supporting Malaysia's ongoing energy transition.

Overall, this review not only synthesises the current state of ESS performance evaluation in Malaysia but also provides an evidence-based, systems-oriented framework to guide future research, policy formulation, and SSCM. By addressing the identified research gaps, the proposed framework offers a practical foundation for strengthening ESS deployment and accelerating Malaysia's national energy transition.

Author Contributions

Conceptualization, S.M.S.; methodology, S.M.S.; formal analysis, S.M.S.; writing—original draft preparation, S.M.S.; writing—review and editing, S.M.S. and E.I.R. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data supporting the findings of this systematic review are derived from the published studies included in the review. The study-level data extraction, indicator coding, and quality and relevance appraisal are provided in Supplementary Tables S1 and S2. No new primary dataset was generated.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors used generative AI tools for language editing, formatting, and improving the clarity of the manuscript. All scientific content, analysis, interpretation, and conclusions were reviewed and verified by the authors.

References

- [1] Sustainable Energy Development Authority (SEDA) Malaysia, “Malaysia renewable energy roadmap,” 2021. https://www.seda.gov.my/reportal/wp-content/uploads/2021/12/MyRER_webVer-1.pdf
- [2] Malaysia Ministry of Economy, “National energy transition roadmap,” 2023. <https://ekonomi.gov.my/sites/default/files/2023-08/National%20Energy%20Transition%20Roadmap.pdf>
- [3] X. Luo, J. Wang, M. Dooner, and J. Clarke, “Overview of current development in electrical energy storage technologies and the application potential in power system operation,” *Appl. Energy*, vol. 137, pp. 511–536, 2015. <https://doi.org/10.1016/j.apenergy.2014.09.081>
- [4] International Renewable Energy Agency (IRENA), “Malaysia energy transition outlook,” 2023. https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2023/Mar/IRENA_Malaysia_energy_transition_outlook_2023.pdf
- [5] B. Zakeri and S. Syri, “Electrical energy storage systems: A comparative life cycle cost analysis,” *Renew. Sustain. Energy Rev.*, vol. 42, pp. 569–596, 2015. <https://doi.org/10.1016/j.rser.2014.10.011>
- [6] N. Kittner, F. Lill, and D. M. Kammen, “Energy storage deployment and innovation for the clean energy transition,” *Nat. Energy*, vol. 2, no. 9, p. 17125, 2017. <https://doi.org/10.1038/nenergy.2017.125>
- [7] M. A. Pellow, H. Ambrose, D. Mulvaney, R. Betita, and S. Shaw, “Research gaps in environmental life cycle assessments of lithium ion batteries for grid-scale stationary energy storage systems: End-of-life options and other issues,” *Sustain. Mater. Technol.*, vol. 23, p. e00120, 2020. <https://doi.org/10.1016/j.susmat.2019.e00120>
- [8] J. Więckowski, B. Kizielewicz, A. Shekhovtsov, and W. Sałabun, “How do the criteria affect sustainable supplier evaluation?—A case study using multi-criteria decision analysis methods in a fuzzy environment,” *J. Eng. Manag. Syst. Eng.*, vol. 2, no. 1, pp. 37–52, 2023. <https://doi.org/10.56578/jemse020102>
- [9] International Renewable Energy Agency (IRENA), “RE-organising power systems for the transition,” 2022. https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2022/Jun/IRENA_Organising_Power_Systems_2022.pdf
- [10] International Energy Agency (IEA), “World energy outlook 2023,” 2023. <https://www.iea.org/reports/world-energy-outlook-2023>
- [11] BloombergNEF, “Lithium-ion battery pack prices hit record low of \$139/kWh,” 2023. <https://about.bnef.com/insights/clean-energy/lithium-ion-battery-pack-prices-hit-record-low-of-139-kwh/>
- [12] Rystad Energy, “Energy storage outlook 2025,” 2025. <https://www.rystadenergy.com/insights/whitepaper-energy-storage-outlook>
- [13] Sustainable Energy Development Authority (SEDA) Malaysia, “Net energy metering (NEM) 3.0.” <https://www.seda.gov.my/reportal/nem/>
- [14] N. Y. Amponsah, M. Trolborg, B. Kington, I. Aalders, and R. L. Hough, “Greenhouse gas emissions from renewable energy sources: A review of lifecycle considerations,” *Renew. Sustain. Energy Rev.*, vol. 39, pp. 461–475, 2014. <https://doi.org/10.1016/j.rser.2014.07.087>
- [15] REN21, “Renewables 2020: Global status report,” 2020. https://www.ren21.net/wp-content/uploads/2019/05/gsr_2020_full_report_en.pdf
- [16] A. S. Mohd Razif, N. F. Ab Aziz, M. Z. A. Ab Kadir, and K. Kamil, “Accelerating energy transition through battery energy storage systems deployment: A review on current status, potential and challenges in Malaysia,” *Energy Strateg. Rev.*, vol. 52, p. 101346, 2024. <https://doi.org/10.1016/j.esr.2024.101346>
- [17] U.S. Department of Energy (DOE), “Grid energy storage supply chain deep dive assessment,” 2022. <https://www.energy.gov/sites/default/files/2022-02/Energy%20Storage%20Supply%20Chain%20Report%20-%20final.pdf>
- [18] P. Muthukumar, A. Kumar, M. Afzal, S. Bhogilla, P. Sharma, A. Parida, S. Jana, E. A. Kumar, R. K. Pai, and I. Jain, “Review on large-scale hydrogen storage systems for better sustainability,” *Int. J. Hydrogen Energy*, vol. 48, no. 85, pp. 33 223–33 259, 2023. <https://doi.org/10.1016/j.ijhydene.2023.04.304>
- [19] N. Mandal, P. Sarkar, and N. Petrović, “Multi-criteria decision-making for ranking renewable energy sources: A case study from the Republic of Serbia,” *J. Eng. Manag. Syst. Eng.*, vol. 4, no. 2, pp. 109–121, 2025. <https://doi.org/10.56578/jemse040203>

- [20] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan *et al.*, “The PRISMA 2020 statement: An updated guideline for reporting systematic reviews,” *BMJ*, vol. 372, p. n71, 2021. <https://doi.org/10.1136/bmj.n71>
- [21] N. I. Ilham, N. Y. Dahlan, and M. Z. Hussin, “Techno-economic viability of energy-sharing model for net-zero emissions in the Malaysian corporate sector,” *Renew. Energy Focus*, vol. 54, p. 100704, 2025. <https://doi.org/10.1016/j.ref.2025.100704>
- [22] M. I. Fernandez, Y. I. Go, M. L. D. Wong, and W. G. Früh, “Modelling and development of utility-scale energy system for Malaysia with advanced energy balance analysis: Supply, demand and trade,” *Renew. Energy*, vol. 247, p. 123013, 2025. <https://doi.org/10.1016/j.renene.2025.123013>
- [23] D. H. Didane, B. Manshoor, A. J. Alimin, and A. A. Amin, “Bibliometric analysis of 50 years of energy research in Malaysia: Trends and opportunities,” *Renew. Sustain. Energy Rev.*, vol. 214, p. 115498, 2025. <https://doi.org/10.1016/j.rser.2025.115498>
- [24] M. Adriana, N. Y. Dahlan, and M. A. S. Abd Aziz, “Assessing techno-economic impacts of hybrid renewable energy system with energy storage for campus buildings in Malaysia using HOMER,” *PaperASIA*, vol. 41, no. 3b, pp. 146–155, 2025. <https://doi.org/10.59953/paperasia.v41i3b.381>
- [25] M. I. Fernandez, Y. I. Go, W. G. Früh, and D. M. L. Wong, “Projection of electricity generation profiles and carbon emissions towards 2050: A Malaysia context,” *Energy Sustain. Dev.*, vol. 85, p. 101681, 2025. <https://doi.org/10.1016/j.esd.2025.101681>
- [26] M. Hasan, Y. S. Lim, L. C. Hau, X. C. Miow, J. Wong, D. H. C. Lam, N. S. Hasan, and H. B. Adlan, “An innovative adaptive threshold-based BESS controller utilizing deep learning forecast for peak demand reductions,” *IEEE Access*, vol. 13, pp. 87 582–87 599, 2025. <https://doi.org/10.1109/ACCESS.2025.3570745>
- [27] M. R. Maghami, J. Pasupuleti, and S. Sundaram, “Impact of time of use program on widely distribution generation interconnection in urban power network,” *J. Energy Storage*, vol. 101, p. 113789, 2024. <https://doi.org/10.1016/j.est.2024.113789>
- [28] M. I. Fernandez, Y. I. Go, M. L. D. Wong, and W. G. Früh, “Malaysia’s energy transition and readiness towards attaining net zero: Review of the potential, constraints, and enablers,” *Renew. Energy Focus*, vol. 51, p. 100640, 2024. <https://doi.org/10.1016/j.ref.2024.100640>
- [29] M. A. Jusoh, M. Z. Ibrahim, and M. Z. Daud, “An improved rule-based control of battery energy storage for hourly power dispatching of photovoltaic sources,” *Int. J. Electr. Comput. Eng.*, vol. 14, no. 4, pp. 3783–3791, 2024. <https://doi.org/10.11591/ijece.v14i4.pp3783-3791>
- [30] M. Roslan, V. K. Ramachandramurthy, M. Mansor, A. Mokhzani, K. P. Jern, R. Begum, and M. Hannan, “Techno-economic impact analysis for renewable energy-based hydrogen storage integrated grid electric vehicle charging stations in different potential locations of Malaysia,” *Energy Strateg. Rev.*, vol. 54, p. 101478, 2024. <https://doi.org/10.1016/j.esr.2024.101478>
- [31] M. P. Nguyen, T. Ponomarenko, and N. Nguyen, “Energy transition in Vietnam: A strategic analysis and forecast,” *Sustainability*, vol. 16, no. 5, p. 1969, 2024. <https://doi.org/10.3390/su16051969>
- [32] W. H. Tee, C. K. Gan, and J. Sardi, “Benefits of energy storage systems and its potential applications in Malaysia: A review,” *Renew. Sustain. Energy Rev.*, vol. 192, p. 114216, 2024. <https://doi.org/10.1016/j.rser.2023.114216>
- [33] I. B. Nor Anwar, M. N. Mohd Hussain, S. Z. Mohammad Noor, N. Y. Dahalan, M. S. Onn, and A. Tumian, “Micro-grid of batteray energy storage system (BESS) design for Malaysia’s net energy metering (NEM),” *E3S Web Conf.*, vol. 473, p. 02001, 2024. <https://doi.org/10.1051/e3sconf/202447302001>
- [34] M. Weber, L. Pressburger, L. W. Chau, Z. Khan, T. Waite, M. I. Westphal, G. H. T. Ling, C. S. Ho, and M. Evans, “Carbon neutrality in Malaysia and Kuala Lumpur: Insights from stakeholder-driven integrated assessment modeling,” *Front. Energy Res.*, vol. 12, p. 1336045, 2024. <https://doi.org/10.3389/fenrg.2024.1336045>
- [35] M. T. Sarker, M. H. S. M. Haram, G. Ramasamy, F. Al Farid, and S. Mansor, “Solar photovoltaic home systems in Malaysia: A comprehensive review and analysis,” *Energies*, vol. 16, no. 23, p. 7718, 2023. <https://doi.org/10.3390/en16237718>
- [36] H. C. Lau, “Decarbonization of ASEAN’s power sector: A holistic approach,” *Energy Rep.*, vol. 9, pp. 676–702, 2023. <https://doi.org/10.1016/j.egyr.2022.11.209>
- [37] J. Y. Lee, A. K. Ramasamy, K. H. Ong, R. Verayiah, H. Mokhlis, and M. Marsadek, “Energy storage systems: A review of its progress and outlook, potential benefits, barriers and solutions within the Malaysian distribution network,” *J. Energy Storage*, vol. 72, p. 108360, 2023. <https://doi.org/10.1016/j.est.2023.108360>
- [38] T. S. Le, T. N. Nguyen, D. K. Bui, and T. D. Ngo, “Optimal sizing of renewable energy storage: A techno-economic analysis of hydrogen, battery and hybrid systems considering degradation and seasonal storage,” *Appl. Energy*, vol. 336, p. 120817, 2023. <https://doi.org/10.1016/j.apenergy.2023.120817>
- [39] S. Sreenath, A. M. Azmi, and Z. A. M. Ismail, “Feasibility of solar hybrid energy system at a conservation park: Technical, economic, environmental analysis,” *Energy Rep.*, vol. 9, pp. 711–719, 2023. <https://doi.org/>

- [40] N. E. M. Zahari, H. Mokhlis, N. N. Mansor, and M. F. Sulaima, "Modelling of hybrid energy system for demand-side management: A case study of industrial customer in Malaysia," in *2023 Innovations in Power and Advanced Computing Technologies (i-PACT)*, Kuala Lumpur, Malaysia, 2023, pp. 1–6. <https://doi.org/10.1109/i-PACT58649.2023.10434432>
- [41] K. A. C. Kamaruddin, N. N. Mansor, N. A. M. I. Yeong, and A. F. Othman, "Optimal integration of large-scale solar and battery energy storage system in Malaysia," in *2023 Innovations in Power and Advanced Computing Technologies (i-PACT)*, Kuala Lumpur, Malaysia, 2023, pp. 1–7. <https://doi.org/10.1109/i-PACT58649.2023.10434585>
- [42] J. Hossain, A. F. A. Kadir, H. Shareef, and M. A. Hossain, "Hybrid PV and battery system sizing for commercial buildings in Malaysia: A case study of FKE-2 building in UTeM," in *2023 IEEE IAS Global Conference on Renewable Energy and Hydrogen Technologies (GlobConHT)*, Male, Maldives, 2023, pp. 1–6. <https://doi.org/10.1109/GlobConHT56829.2023.10087792>
- [43] K. Citalingam and Y. I. Go, "Hybrid energy storage design and dispatch strategy evaluation with sensitivity analysis: Techno-economic-environmental assessment," *Energy Storage*, vol. 4, no. 5, p. e353, 2022. <https://doi.org/10.1002/est2.353>
- [44] A. K. Pandey, B. Kalidasan, R. R. Kumar, S. Rahman, V. V. Tyagi, Krismadinata, Z. Said, P. A. Salam, D. E. Juanico, J. U. Ahamed *et al.*, "Solar energy utilization techniques, policies, potentials, progresses, challenges and recommendations in ASEAN countries," *Sustainability*, vol. 14, no. 18, p. 11193, 2022. <https://doi.org/10.3390/su141811193>
- [45] L. M. A. Al-Hilfi, S. Morris, A. P. Fathima, and M. Ezra, "Investigation of potential benefits and challenges of using gravity energy storage in residential sectors," in *2022 International Virtual Conference on Power Engineering Computing and Control: Developments in Electric Vehicles and Energy Sector for Sustainable Future (PECCON)*, Chennai, India, 2022, pp. 1–6. <https://doi.org/10.1109/PECCON55017.2022.9851072>
- [46] C. J. Weng, S. Y. Yip, W. K. Muzammil, and M. A. Ismail, "Standalone solar power generation with dynamic error-driven PI-based energy management system for rural electrification in Malaysia," in *2022 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS)*, Shah Alam, Malaysia, 2022, pp. 66–71. <https://doi.org/10.1109/I2CACIS54679.2022.9815493>
- [47] L. M. A. Al-Hilfi, S. Morris, J. Wong, B. Venkatesh, K. W. Chew, and K. H. Chua, "Exploration of the suitability of gravity energy storage in Malaysian grid systems," in *2022 IEEE International Conference in Power Engineering Application (ICPEA)*, Shah Alam, Malaysia, 2022, pp. 1–6. <https://doi.org/10.1109/ICPEA53519.2022.9744647>
- [48] A. Halim, A. Fudholi, K. Sopian, and S. J. Phillips, "Performance of hybrid solar photovoltaic–diesel generator and battery storage design for rural electrification in Malaysia," *Int. J. Sustain. Dev. Plan.*, vol. 16, no. 5, pp. 883–893, 2021. <https://doi.org/10.18280/ijstdp.160509>
- [49] M. Beuse, M. Dirksmeier, B. Steffen, and T. S. Schmidt, "Profitability of commercial and industrial photovoltaics and battery projects in South-East-Asia," *Appl. Energy*, vol. 271, p. 115218, 2020. <https://doi.org/10.1016/j.apenergy.2020.115218>
- [50] L. H. Saw, J. S. Cheow, Y. J. King, W. H. Yeo, B. H. Lim, J. Y. Tey, and T. C. Ng, "Numerical modelling of the power pack system for remote islands electrification," *IOP Conf. Ser. Earth Environ. Sci.*, vol. 268, no. 1, p. 012103, 2019. <https://doi.org/10.1088/1755-1315/268/1/012103>
- [51] W. Jing, C. H. Lai, D. K. X. Ling, W. S. H. Wong, and M. L. D. Wong, "Battery lifetime enhancement via smart hybrid energy storage plug-in module in standalone photovoltaic power system," *J. Energy Storage*, vol. 21, pp. 586–598, 2019. <https://doi.org/10.1016/j.est.2018.12.007>
- [52] M. Shamshiri, C. K. Gan, N. I. Zolkifri, and K. A. Baharin, "Battery energy storage system sizing for high penetration of solar photovoltaic systems in low voltage distribution network," *ARNP J. Eng. Appl. Sci.*, vol. 13, no. 19, pp. 8103–8109, 2018.
- [53] D. F. Dominković, V. Dobravec, Y. Jiang, P. S. Nielsen, and G. Krajačić, "Modelling smart energy systems in tropical regions," *Energy*, vol. 155, pp. 592–609, 2018. <https://doi.org/10.1016/j.energy.2018.05.007>
- [54] N. D. Nordin and H. A. Rahman, "Sizing and economic analysis of stand alone photovoltaic system with hydrogen storage," *IOP Conf. Ser. Earth Environ. Sci.*, vol. 93, p. 012068, 2017. <https://doi.org/10.1088/1755-1315/93/1/012068>
- [55] T. Kamal, M. Karabacak, S. Z. Hassan, H. Li, A. Arsalan, and R. K. Rajkumar, "Integration and control of an off-grid hybrid wind/PV generation system for rural applications," in *2017 International Conference on Innovations in Electrical Engineering and Computational Technologies (ICIEECT)*, Karachi, Pakistan, 2017, pp. 1–6. <https://doi.org/10.1109/ICIEECT.2017.7916543>
- [56] Y. Wang, B. F. Wang, and P. L. So, "A voltage regulation method using distributed energy storage systems in

- LV distribution networks,” in *2016 IEEE International Energy Conference (ENERGYCON)*, Leuven, Belgium, 2016, pp. 1–6. <https://doi.org/10.1109/ENERGYCON.2016.7514050>
- [57] L. M. Halabi, S. Mekhilef, L. Olatomiwa, and J. Hazelton, “Performance analysis of hybrid PV/diesel/battery system using HOMER: A case study Sabah, Malaysia,” *Energy Convers. Manag.*, vol. 144, pp. 322–339, 2017. <https://doi.org/10.1016/j.enconman.2017.04.070>
- [58] S. Koohi-Kamali, N. A. Rahim, and H. Mokhlis, “Smart power management algorithm in microgrid consisting of photovoltaic, diesel, and battery storage plants considering variations in sunlight, temperature, and load,” *Energy Convers. Manag.*, vol. 84, pp. 562–582, 2014. <https://doi.org/10.1016/j.enconman.2014.04.072>
- [59] M. Z. Ibrahim, K. Sopian, and W. R. W. Daud, “PV-wind-battery hybrid system for hydrogen production in tropical climatic condition through water electrolysis: A case study for Terengganu state, Malaysia,” *Int. Energy J.*, vol. 10, no. 1, pp. 11–18, 2009.